

# Communities and Crime Analysis Dashboard

## Project Overview and Objective:

This project focuses on cleaning, transforming, analysing, and visualizing Communities and Crime data using Excel, Power Query, and Power BI.

The dataset contains population, demographic, socioeconomic, housing, police, and crime-related attributes for different communities. The project aims to transform the raw dataset into a structured and analysis-ready format and develop an interactive Power BI dashboard.

## Main Objectives

- To clean and preprocess raw community and crime data using Excel and Power Query.
- To identify and handle missing values, duplicates, inconsistent data types, and formatting issues.
- To analyse demographic, socioeconomic, housing, police, and crime-related factors.
- To compare crime levels across different communities and states.
- To identify factors that are associated with higher or lower crime levels.
- To create an interactive Power BI dashboard for data-driven analysis and decision-making.

## Data Sources:

Source Description and Timeline

**Dataset:** Communities and Crime Unnormalized Dataset

**Source:** UCI Machine Learning Repository

**Dataset Size:** Approximately 2,215 communities and 147 attributes

**Data Period:** Primarily based on 1990 U.S. Census data and 1995 FBI crime data.

**Geographical Coverage:** Communities across the United States.

## Domain

Crime Analytics / Social Analytics / Public Safety

## Problem Statement:

Communities have different population, demographic, social, economic, housing, law-enforcement, and crime-related characteristics, making it difficult to identify meaningful crime patterns from raw data.

This project aims to clean and transform the Communities and Crime dataset using Excel and Power Query, analyse relationships between community characteristics and crime levels, and develop an interactive Power BI dashboard to compare communities and identify factors associated with crime.

## Key Analysis Questions

1. Which states or communities have higher crime levels?
2. What demographic and socioeconomic characteristics are associated with crime?
3. How do poverty and unemployment relate to crime levels?
4. How do population density and urbanization differ across communities?
5. How do police-related factors vary between communities?
6. Which communities have higher violent-crime rates?
7. What patterns can be identified between socioeconomic conditions and crime?

## Attribute (Column / Features) Details

| Attribute Name | Data Type | Description                             |
|----------------|-----------|-----------------------------------------|
| communityname  | Text      | Name of the community                   |
| State          | Text      | State in which the community is located |
| countyCode     | Integer   | County identification code              |
| communityCode  | Integer   | Community identification code           |

| Attribute Name  | Data Type | Description                                                    |
|-----------------|-----------|----------------------------------------------------------------|
| fold            | Integer   | Data fold used for validation                                  |
| pop             | Numeric   | Population of the community                                    |
| perHoush        | Decimal   | Average number of persons per household                        |
| pctBlack        | Decimal   | Percentage of population that is Black                         |
| pctWhite        | Decimal   | Percentage of population that is White                         |
| pctAsian        | Decimal   | Percentage of population that is Asian                         |
| pctHisp         | Decimal   | Percentage of population that is Hispanic                      |
| pct12-21        | Decimal   | Percentage of population aged 12–21                            |
| pct12-29        | Decimal   | Percentage of population aged 12–29                            |
| pct16-24        | Decimal   | Percentage of population aged 16–24                            |
| pct65up         | Decimal   | Percentage of population aged 65 and above                     |
| persUrban       | Numeric   | Number of people living in urban areas                         |
| pctUrban        | Decimal   | Percentage of population living in urban areas                 |
| medIncome       | Numeric   | Median household income                                        |
| pctWwage        | Decimal   | Percentage of households receiving wage/salary income          |
| pctWfarm        | Decimal   | Percentage of households receiving farm/self-employment income |
| pctPoverty      | Decimal   | Percentage of population below the poverty level               |
| pctUnemploy     | Decimal   | Percentage of unemployed population                            |
| pctCollGrad     | Decimal   | Percentage of population who are college graduates             |
| popDensity      | Decimal   | Population density                                             |
| murders         | Numeric   | Number of murders                                              |
| murdPerPop      | Decimal   | Murder rate per population                                     |
| rapes           | Numeric   | Number of rapes                                                |
| rapesPerPop     | Decimal   | Rape rate per population                                       |
| robberies       | Numeric   | Number of robberies                                            |
| robberPerPop    | Decimal   | Robbery rate per population                                    |
| assaults        | Numeric   | Number of assaults                                             |
| assaultPerPop   | Decimal   | Assault rate per population                                    |
| burglaries      | Numeric   | Number of burglaries                                           |
| burglPerPop     | Decimal   | Burglary rate per population                                   |
| larcenies       | Numeric   | Number of larcenies                                            |
| larcPerPop      | Decimal   | Larceny rate per population                                    |
| autoTheft       | Numeric   | Number of auto thefts                                          |
| autoTheftPerPop | Decimal   | Auto-theft rate per population                                 |
| arsons          | Numeric   | Number of arsons                                               |
| arsonsPerPop    | Decimal   | Arson rate per population                                      |
| violentPerPop   | Decimal   | Violent crime rate per population                              |
| nonViolPerPop   | Decimal   | Non-violent crime rate per population                          |

## Tools & Technologies

### Excel

Used for:

- Initial data inspection
- Data organization

- Filtering and sorting
- Duplicate checking
- Basic data validation
- Pivot Tables
- Initial exploratory analysis

## **Power Query**

Used for:

- Data cleaning
- Handling missing values
- Changing data types
- Renaming columns
- Removing unnecessary columns
- Standardizing data
- Creating transformation steps
- Preparing the dataset for Power BI

## **Power BI**

Used for:

- Data modelling
- Relationships between tables
- DAX calculations
- Interactive visualizations
- Slicers and filters
- Drill-down analysis
- Dashboard creation
- Crime and community analysis

## **DAX**

Used for calculating important metrics such as:

- Total Population
- Average Population
- Total Murders
- Total Robberies
- Total Assaults
- Average Violent Crime Rate
- Average Non-Violent Crime Rate
- Crime Rate comparisons

- State/community rankings

## **Data Pre-Processing (Excel / Power Query)**

### **Step 1: Fix headers**

The original file had no usable column headers. The first row of real column names was promoted to be the table header so every column had a readable, correct name (e.g. larcPerPop).

### **Step 2: Remove columns with very high missing data**

22 columns were removed because they were missing in approximately 84.47% of rows. Columns missing that much data cannot be reliably imputed or analyzed, so they were dropped rather than filled in.

### **Step 3: Remove non-predictive identifier columns**

countyCode and communityCode were removed. Both were about 55% missing, and even where present they are just ID numbers (not characteristics of a community), so they add no analytical value and were safe to drop.

Columns remaining after Steps 2-3: 123 (down from 147).

### **Step 4: Impute the two main target columns**

ViolentCrimesPerPop and nonViolPerPop (the two key crime-rate columns) had missing values filled in using median imputation - each blank was replaced with the median value of that column. Median was used instead of average because crime rates are skewed by a small number of very high-crime communities, and the median is not distorted by those outliers.

### **Step 5: Check for duplicate rows**

The full table was checked for exact duplicate rows, and also for duplicate combinations of communityname and State (the natural identifier for a community). Result: 0 duplicates found either way, so no rows were removed.

### **Step 6: Check and correct data types**

Every column was reviewed and assigned the correct type:

- 37 columns were set to whole numbers (counts, e.g. murders, pop, medIncome)
- 84 columns were set to decimal numbers (rates and percentages, e.g. pctBlack, murdPerPop)
- communityname and State were set to text, with State standardized to uppercase 2-letter codes

### Step 7: Handle remaining missing values

After Step 4, 15 columns still had gaps - mainly rarer crime types such as rapes, arsons, and their per-population rates, where some small communities simply had zero reported incidents on record. These were filled using the same median imputation method already used in Step 4, keeping the approach consistent across the whole dataset. After this step, 0 missing values remained anywhere in the dataset.

### Step 8: Standardize and clean

- Trimmed extra spaces from community names
- Converted all State codes to uppercase
- Rounded percentage columns to 4 decimal places for consistency
- Confirmed all State codes are valid 2-letter US codes (48 states/DC present in this dataset)

### Step 9: Filter and sort

Rows were checked for any blank community name or state (none were found, so no rows needed to be removed). The table was then sorted alphabetically by State, then by community name, to make it easier to browse and to give a predictable, repeatable order.

### Step 10: Create calculated fields

Several new columns were added to make analysis easier:

| New column              | How it's calculated                         |
|-------------------------|---------------------------------------------|
| TotalViolentCrimeCount  | murders + rapes + robberies + assaults      |
| TotalPropertyCrimeCount | burglaries + larcenies + autoTheft + arsons |

| New column             | How it's calculated                                                       |
|------------------------|---------------------------------------------------------------------------|
| TotalCrimeCount        | TotalViolentCrimeCount + TotalPropertyCrimeCount                          |
| PctMinorityPop         | pctBlack + pctAsian + pctHisp                                             |
| CrimeSeverityIndex     | (ViolentCrimesPerPop x 2) + nonViolPerPop, weighting violent crime higher |
| PopulationSizeCategory | pop split into Small / Medium / Large / Very Large using quartiles        |

### Step 11: Build Fact and Dimension tables

To make the dataset Power BI-friendly, it was split into a simple star schema, all linked by a new CommunityKey column:

- Dim\_Community - location and core demographic/economic fields (population, income, poverty, land area, etc.)
- Fact\_Crime - all crime counts, crime rates, and the new calculated crime fields
- Dim\_Socioeconomic - the remaining detailed demographic, housing, and economic columns

### Step 12: Validate the final dataset

Before delivering the final file, seven checks were run to confirm the cleaning worked correctly:

| Check                                                  | Result |
|--------------------------------------------------------|--------|
| Row count still matches original (2,215)               | Passed |
| No missing values remain                               | Passed |
| No duplicate CommunityKey values                       | Passed |
| No duplicate communityname + State pairs               | Passed |
| All percentage columns fall between 0 and 100          | Passed |
| All State codes are 2 letters                          | Passed |
| Fact table row count matches Dimension table row count | Passed |

### Step 13: Save each stage separately

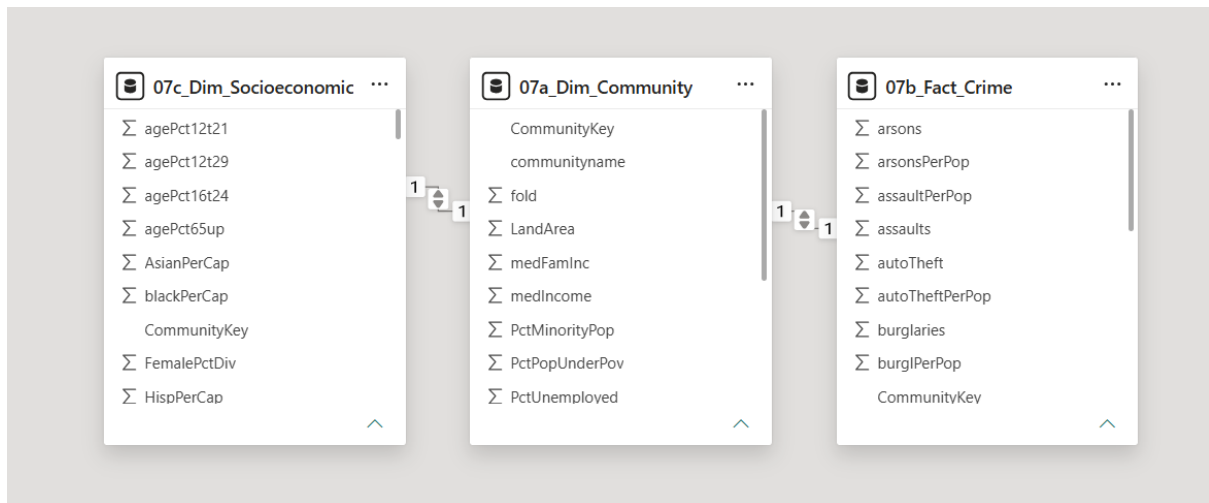
Every major stage was saved as its own sheet in the workbook, rather than overwriting previous work. This keeps a full audit trail and means no earlier work is ever lost:

| Sheet name                 | Contents                                                         |
|----------------------------|------------------------------------------------------------------|
| Raw_Data                   | Original, completely unedited file (preserved as-is)             |
| Headers                    | Raw data with headers fixed only                                 |
| Handle Missing Values      | Working copy after Steps 2-4 (column removal + first imputation) |
| 01_Duplicate_Check         | Duplicate check results                                          |
| 02_Data_Types              | Record of which columns were set to which data type              |
| 03_Missing_Values_Handled  | Log of every column imputed and the median value used            |
| 04_Standardized            | Notes on cleanup actions taken                                   |
| 05_Filtered_Sorted         | Filter/sort summary                                              |
| 06_Calculated_Fields       | Documentation of every new formula/column                        |
| 07a / 07b / 07c            | Dim_Community, Fact_Crime, Dim_Socioeconomic tables              |
| 08_Validation              | Results of the 7 validation checks                               |
| 09_Final_Cleaned_FlatTable | The final, single flat table - ready for analysis                |



## 7. Data Modelling and DAX (Power BI)

### Relationships



### Create DAX Measures

- **Measure 1 — Total Communities**

The screenshot shows the Power BI interface with the following details:

- Formula Bar**:

```
1 Total Communities =  
2 DISTINCTCOUNT('07a_Dim_Community'[CommunityKey])
```
- Properties Pane**:
  - Name**: Total Communities
  - Home table**: 07b\_Fact\_Crime
  - Description**: Enter a description
  - Synonyms**: total communities, communities
  - Display folder**: (empty)
- Data Pane**: Shows a list of tables and measures, including 07c\_Dim\_Socioeconomic, 07a\_Dim\_Community, and 07b\_Fact\_Crime.

- **Measure 2 — Average Violent Crime**

The screenshot shows the Power BI interface with the following details:

- Formula Bar**:

```
1 Average Violent Crime =  
2 AVERAGE('07b_Fact_Crime'[ViolentCrimesPerPop])
```
- Properties Pane**:
  - Name**: Average Violent Crime
  - Home table**: 07b\_Fact\_Crime
  - Description**: Enter a description
  - Synonyms**: (empty)
  - Display folder**: (empty)
- Data Pane**: Shows a list of tables and measures, including 07c\_Dim\_Socioeconomic, 07a\_Dim\_Community, and 07b\_Fact\_Crime.

### • Measure 3 — Average Non-Violent Crime

The screenshot shows the Power BI Desktop interface. The formula bar at the top contains the following DAX formula:

```
1 Average NonViolent Crime =
2 AVERAGE('07b_Fact_Crime'[nonViolPerPop])
```

The data model is visualized in the background, showing three tables: 07c\_Dim\_Socioeco..., 07a\_Dim\_Community, and 07b\_Fact\_Crime. The 07b\_Fact\_Crime table is highlighted in the Properties pane on the right. The Data pane on the right shows a list of tables, including Average NonViolent Crime, autoTheft, autoTheftPerPop, burglaries, burgPerPop, CommunityKey, CrimeSeverityIndex, larcenies, larcPerPop, murders, murdPerPop, and nonViolPerPop.

### • Measure 4 — Total Murders

The screenshot shows the Power BI Desktop interface. The formula bar at the top contains the following DAX formula:

```
1 Total Murders =
2 SUM('07b_Fact_Crime'[murders])
```

The data model is visualized in the background, showing three tables: 07c\_Dim\_Socioeconomic, 07a\_Dim\_Community, and 07b\_Fact\_Crime. The 07b\_Fact\_Crime table is highlighted in the Properties pane on the right. The Data pane on the right shows a list of tables, including Average Violent Crime, burglaries, burgPerPop, CommunityKey, CrimeSeverityIndex, larcenies, larcPerPop, murders, murdPerPop, nonViolPerPop, rapes, rapesPerPop, robbPerPop, robberies, Total Communities, Total Murders, TotalCrimeCount, and TotalTheftCrimeCount.

### • Measure 5 — Total Robberies

The screenshot shows the Power BI Desktop interface. The formula bar at the top contains the following DAX formula:

```
1 Total Robberies =
2 SUM('07b_Fact_Crime'[robberies])
```

The data model is visualized in the background, showing three tables: 07c\_Dim\_Socioeco..., 07a\_Dim\_Community, and 07b\_Fact\_Crime. The 07b\_Fact\_Crime table is highlighted in the Properties pane on the right. The Data pane on the right shows a list of tables, including Average Violent Crime, burglaries, burgPerPop, CommunityKey, CrimeSeverityIndex, larcenies, larcPerPop, murders, murdPerPop, nonViolPerPop, rapes, rapesPerPop, robbPerPop, robberies, Total Communities, Total Murders, TotalCrimeCount, and TotalTheftCrimeCount.

## • Measure 6 — Total Assaults

1 Total Assaults =  
2 SUM('07b\_Fact\_Crime'[assaults])

**07c\_Dim\_Socioeconomic**

- Σ agePct12t21
- Σ agePct12t29
- Σ agePct16t24
- Σ agePct65up
- Σ AsianPerCap
- Σ blackPerCap
- CommunityKey
- Σ FemalePctDiv
- Σ HispPerCap

**07a\_Dim\_Community**

- Σ medFamInc
- Σ medIncInc
- Σ PctMinorityPop
- Σ PctPopUnderPov
- Σ PctUnemployed
- Σ perCapInc
- Σ perHoush
- Σ pop
- DimName

**07b\_Fact\_Crime**

- Σ arsons
- Σ arsonsPerPop
- Σ assaultPerPop
- Σ assaults
- Σ autoTheft
- Σ autoTheftPerPop
- Σ burglaries
- Σ burgPerPop
- CommunityKey

**Properties**

Enter a description

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**Synonyms**

total assaults, assaults

**Display folder**

Enter the display folder

**Is hidden**

No

**Data**

**Tables** Model

Search

- Average Violent Crime
- Σ burglaries
- Σ burgPerPop
- CommunityKey
- Σ CrimeSeverityIndex
- Σ larcenies
- Σ larcPerPop
- Σ murders
- Σ murdPerPop
- Σ nonViolPerPop
- Σ rapes
- Σ rapesPerPop
- Σ robbPerPop
- Σ robberies
- Total Assaults**
- Total Communities

## • Measure 7 — Total Burglaries

1 Total Burglaries =  
2 SUM('07b\_Fact\_Crime'[burglaries])

**07c\_Dim\_Socioeconomic**

- Σ agePct12t21
- Σ agePct12t29
- Σ agePct16t24
- Σ agePct65up
- Σ AsianPerCap
- Σ blackPerCap
- CommunityKey
- Σ FemalePctDiv
- Σ HispPerCap

**07a\_Dim\_Community**

- Σ medFamInc
- Σ medIncInc
- Σ PctMinorityPop
- Σ PctPopUnderPov
- Σ PctUnemployed
- Σ perCapInc
- Σ perHoush
- Σ pop
- DimName

**07b\_Fact\_Crime**

- Σ arsons
- Σ arsonsPerPop
- Σ assaultPerPop
- Σ assaults
- Σ autoTheft
- Σ autoTheftPerPop
- Σ burglaries
- Σ burgPerPop
- CommunityKey

**Properties**

Enter a description

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**Synonyms**

total burglaries, burglaries

**Display folder**

Enter the display folder

**Is hidden**

No

**Data**

**Tables** Model

Search

- Average Violent Crime
- Σ burglaries
- Σ burgPerPop
- CommunityKey
- Σ CrimeSeverityIndex
- Σ larcenies
- Σ larcPerPop
- Σ murders
- Σ murdPerPop
- Σ nonViolPerPop
- Σ rapes
- Σ rapesPerPop
- Σ robbPerPop
- Σ robberies
- Total Assaults**
- Total Burglaries**
- Total Communities

## • Measure 8 — Total Auto Theft

1 Total Auto Theft =  
2 SUM('07b\_Fact\_Crime'[autoTheft])

**07c\_Dim\_Socioeconomic**

- Σ agePct12t21
- Σ agePct12t29
- Σ agePct16t24
- Σ agePct65up
- Σ AsianPerCap
- Σ blackPerCap
- CommunityKey
- Σ FemalePctDiv
- Σ HispPerCap

**07a\_Dim\_Community**

- Σ medFamInc
- Σ medIncInc
- Σ PctMinorityPop
- Σ PctPopUnderPov
- Σ PctUnemployed
- Σ perCapInc
- Σ perHoush
- Σ pop
- DimName

**07b\_Fact\_Crime**

- Σ arsons
- Σ arsonsPerPop
- Σ assaultPerPop
- Σ assaults
- Σ autoTheft
- Σ autoTheftPerPop
- Σ burglaries
- Σ burgPerPop
- CommunityKey

**Properties**

Enter a description

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**Synonyms**

total auto theft, theft

**Display folder**

Enter the display folder

**Is hidden**

No

**Data**

**Tables** Model

Search

- Average Violent Crime
- Σ burglaries
- Σ burgPerPop
- CommunityKey
- Σ CrimeSeverityIndex
- Σ larcenies
- Σ larcPerPop
- Σ murders
- Σ murdPerPop
- Σ nonViolPerPop
- Σ rapes
- Σ rapesPerPop
- Σ robbPerPop
- Σ robberies
- Total Assaults**
- Total Auto Theft**
- Total Burglaries

- **Measure 9 — Total Arsons**

1 Total Arsons =  
2 SUM('07b\_Fact\_Crime'[arsons])

**07c\_Dim\_Socioeconomic**

- Σ agePct12t21
- Σ agePct12t29
- Σ agePct16t24
- Σ agePct65up
- Σ AsianPerCap
- Σ blackPerCap
- CommunityKey
- Σ FemalePctDiv
- Σ HispPerCap

**07a\_Dim\_Community**

- Σ medFamInc
- Σ medIncome
- Σ PctMinorityPop
- Σ PctPopUnderPov
- Σ PctUnemployed
- Σ perCapInc
- Σ perHoush
- Σ pop
- CommunityKey

**07b\_Fact\_Crime**

- Σ arsons
- Σ arsonsPerPop
- Σ assaultPerPop
- Σ assaults
- Σ autoTheft
- Σ autoTheftPerPop
- Σ burglaries
- Σ burglaryPerPop
- CommunityKey

**Properties**

Enter a description

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**Synonyms**

total arsons, arsons

**Display folder**

Enter the display folder

**Is hidden**

☐ No

**Data**

**Tables** Model

Search

- Average Violent Crime
- Σ burglaries
- Σ burglaryPerPop
- CommunityKey
- Σ CrimeSeverityIndex
- Σ larcenies
- Σ larcPerPop
- Σ murders
- Σ murderPerPop
- Σ nonViolPerPop
- Σ rapes
- Σ rapesPerPop
- Σ robbPerPop
- Σ robberies
- Total Arsons**

## Create Community Measures

- **Total Communities**

1 Total Communities =  
2 DISTINCTCOUNT('07a\_Dim\_Community'[CommunityKey])

**07c\_Dim\_Socioeconomic**

- Σ agePct12t21
- Σ agePct12t29
- Σ agePct16t24
- Σ agePct65up
- Σ AsianPerCap
- Σ blackPerCap
- CommunityKey
- Σ FemalePctDiv
- Σ HispPerCap

**07a\_Dim\_Community**

- CommunityKey
- communityname
- Σ fold
- Σ LandArea
- Σ medFamInc
- Σ medIncome
- Σ PctMinorityPop
- Σ PctPopUnderPov
- Σ PctUnemployed

**07b\_Fact\_Crime**

- Σ arsons
- Σ arsonsPerPop
- Σ assaultPerPop
- Σ assaults
- Σ autoTheft
- Σ autoTheftPerPop
- Σ burglaries
- Σ burglaryPerPop
- CommunityKey

**Properties**

**General**

**Name**

Total Communities

**Home table**

07a\_Dim\_Community

**Description**

Enter a description

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**Synonyms**

total communities, communities

**Data**

**Tables** Model

Search

- 07a\_Dim\_Community
- CommunityKey
- communityname
- Σ fold
- Σ LandArea
- Σ medFamInc
- Σ medIncome
- Σ PctMinorityPop
- Σ PctPopUnderPov
- Σ PctUnemployed
- Σ perCapInc
- Σ perHoush
- Σ pop
- Σ PopDens
- PopulationSizeCategory
- State
- Total Communities**

- **Average Poverty**

1 Average Poverty =  
2 AVERAGE('07a\_Dim\_Community'[PctPopUnderPov])

**07c\_Dim\_Socioeconomic**

- Σ agePct12t21
- Σ agePct12t29
- Σ agePct16t24
- Σ agePct65up
- Σ AsianPerCap
- Σ blackPerCap
- CommunityKey
- Σ FemalePctDiv
- Σ HispPerCap

**07a\_Dim\_Community**

- CommunityKey
- communityname
- Σ fold
- Σ LandArea
- Σ medFamInc
- Σ medIncome
- Σ PctMinorityPop
- Σ PctPopUnderPov
- Σ PctUnemployed

**07b\_Fact\_Crime**

- Σ arsons
- Σ arsonsPerPop
- Σ assaultPerPop
- Σ assaults
- Σ autoTheft
- Σ autoTheftPerPop
- Σ burglaries
- Σ burglaryPerPop
- CommunityKey

**Properties**

**General**

**Name**

Average Poverty

**Home table**

07a\_Dim\_Community

**Description**

Enter a description

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**Data**

**Tables** Model

Search

- 07a\_Dim\_Community
- Average Poverty**
- CommunityKey
- communityname
- Σ fold
- Σ LandArea
- Σ medFamInc
- Σ medIncome
- Σ PctMinorityPop
- Σ PctPopUnderPov
- Σ PctUnemployed

- **Average Unemployment**

1 Average Unemployment =  
2 AVERAGE('07a\_Dim\_Community'[PctUnemployed])

**07c\_Dim\_Socioeconomic**

- Σ agePct12t21
- Σ agePct12t29
- Σ agePct16t24
- Σ agePct65up
- Σ AsianPerCap
- Σ blackPerCap
- CommunityKey
- Σ FemalePctDiv
- Σ HispPerCap

**07a\_Dim\_Community**

- CommunityKey
- communityname
- Σ fold
- Σ LandArea
- Σ medFamInc
- Σ medIncome
- Σ PctMinorityPop
- Σ PctPopUnderPov
- Σ PctUnemployed

**07b\_Fact\_Crime**

- Σ arsons
- Σ arsonsPerPop
- Σ assaultPerPop
- Σ assaults
- Σ autoTheft
- Σ autoTheftPerPop
- Σ burglaries
- Σ burglPerPop
- CommunityKey

**Properties**

Name: Average Unemployment

Home table: 07a\_Dim\_Community

Description: Enter a description

Create with Copilot

**Data**

Tables

Search

- 07a\_Dim\_Community
- Average Poverty
- Average Unemployment**
- CommunityKey
- communityname
- Σ fold
- Σ LandArea
- Σ medFamInc
- Σ medIncome
- Σ PctMinorityPop
- Σ PctPopUnderPov

## Create Socioeconomic Measures

- **Average Black Population %**

1 Average Black Population =  
2 AVERAGE('07c\_Dim\_Socioeconomic'[pctBlack])

**07c\_Dim\_Socioeconomic**

- Σ agePct12t21
- Σ agePct12t29
- Σ agePct16t24
- Σ agePct65up
- Σ AsianPerCap
- Σ blackPerCap
- CommunityKey
- Σ FemalePctDiv
- Σ HispPerCap

**07a\_Dim\_Community**

- CommunityKey
- communityname
- Σ fold
- Σ LandArea
- Σ medFamInc
- Σ medIncome
- Σ PctMinorityPop
- Σ PctPopUnderPov
- Σ PctUnemployed

**07b\_Fact\_Crime**

- Σ arsons
- Σ arsonsPerPop
- Σ assaultPerPop
- Σ assaults
- Σ autoTheft
- Σ autoTheftPerPop
- Σ burglaries
- Σ burglPerPop
- CommunityKey

**Properties**

Name: Average Black Population

Home table: 07c\_Dim\_Socioecon

Description: Enter a description

Create with Copilot

**Data**

Tables

Search

- ViolentCrimesPerPop
- 07c\_Dim\_Socioeconomic
- Σ agePct12t21
- Σ agePct12t29
- Σ agePct16t24
- Σ agePct65up
- Σ AsianPerCap
- Average Black Population**
- blackPerCap
- CommunityKey
- Σ FemalePctDiv

- **Average White Population**

1 Average White Population =  
2 AVERAGE('07c\_Dim\_Socioeconomic'[pctWhite])

**07c\_Dim\_Socioeconomic**

- Σ agePct12t21
- Σ agePct12t29
- Σ agePct16t24
- Σ agePct65up
- Σ AsianPerCap
- Σ blackPerCap
- CommunityKey
- Σ FemalePctDiv
- Σ HispPerCap

**07a\_Dim\_Community**

- CommunityKey
- communityname
- Σ fold
- Σ LandArea
- Σ medFamInc
- Σ medIncome
- Σ PctMinorityPop
- Σ PctPopUnderPov
- Σ PctUnemployed

**07b\_Fact\_Crime**

- Σ arsons
- Σ arsonsPerPop
- Σ assaultPerPop
- Σ assaults
- Σ autoTheft
- Σ autoTheftPerPop
- Σ burglaries
- Σ burglPerPop
- CommunityKey

**Properties**

Name: Average White Population

Home table: 07c\_Dim\_Socioecon

Description: Enter a description

Create with Copilot

**Data**

Tables

Search

- ViolentCrimesPerPop
- 07c\_Dim\_Socioeconomic
- Σ agePct12t21
- Σ agePct12t29
- Σ agePct16t24
- Σ agePct65up
- Σ AsianPerCap
- Average Black Population
- Average White Population**
- blackPerCap
- CommunityKey

## 8. Analysis and Visualizations (Power BI)

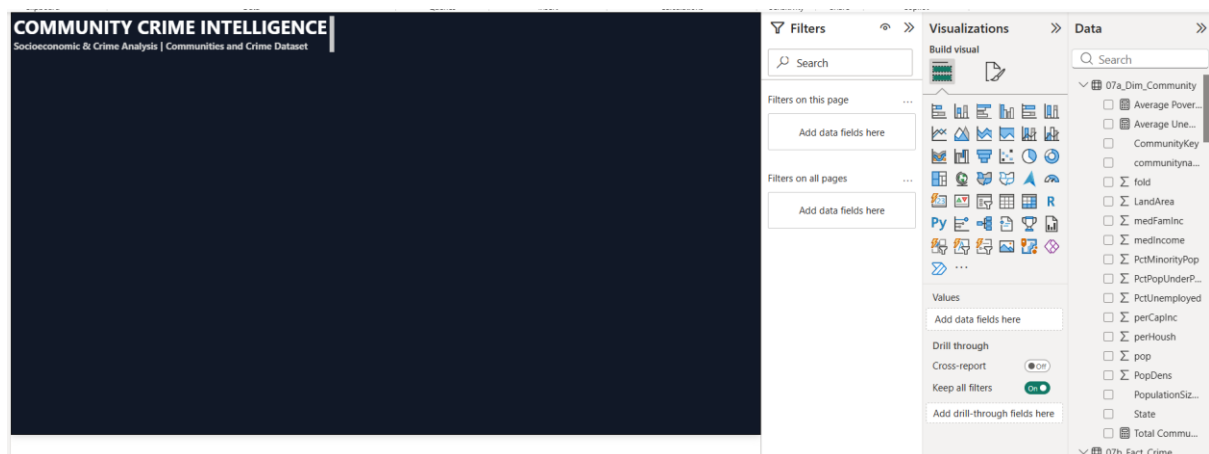
### Step 1 — Set the dashboard background

Go to Report View.

Then:

Format → Canvas settings / Page → Background

Set a dark navy/charcoal background.

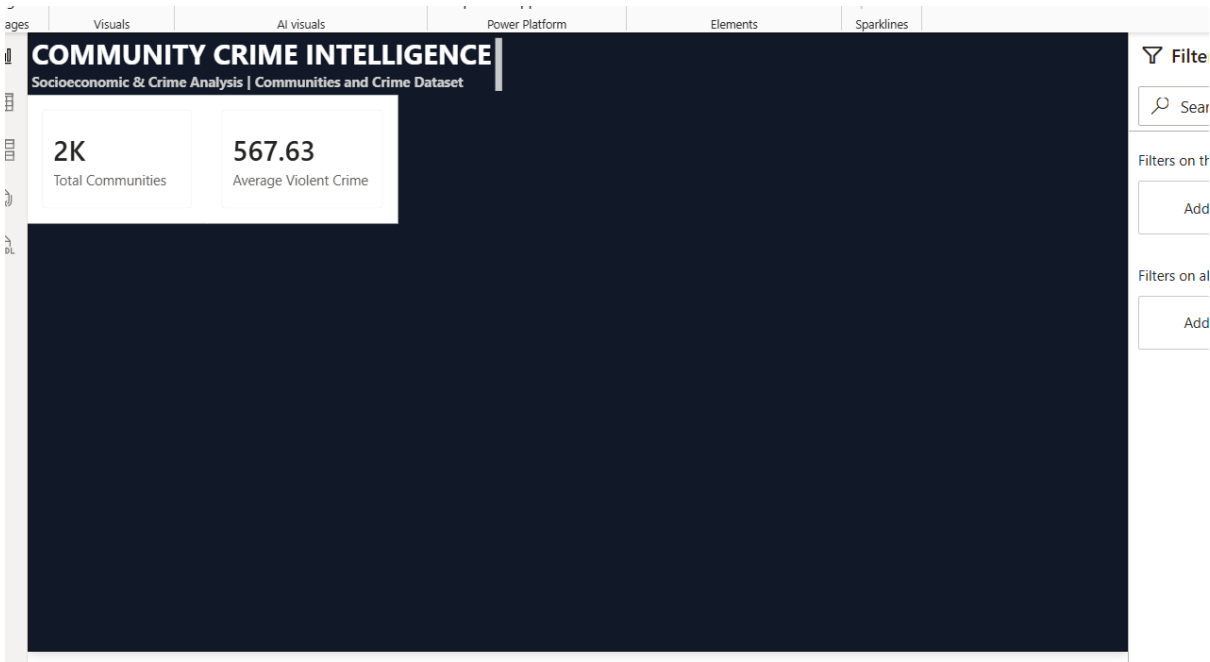


### Step 2 — First KPI Card

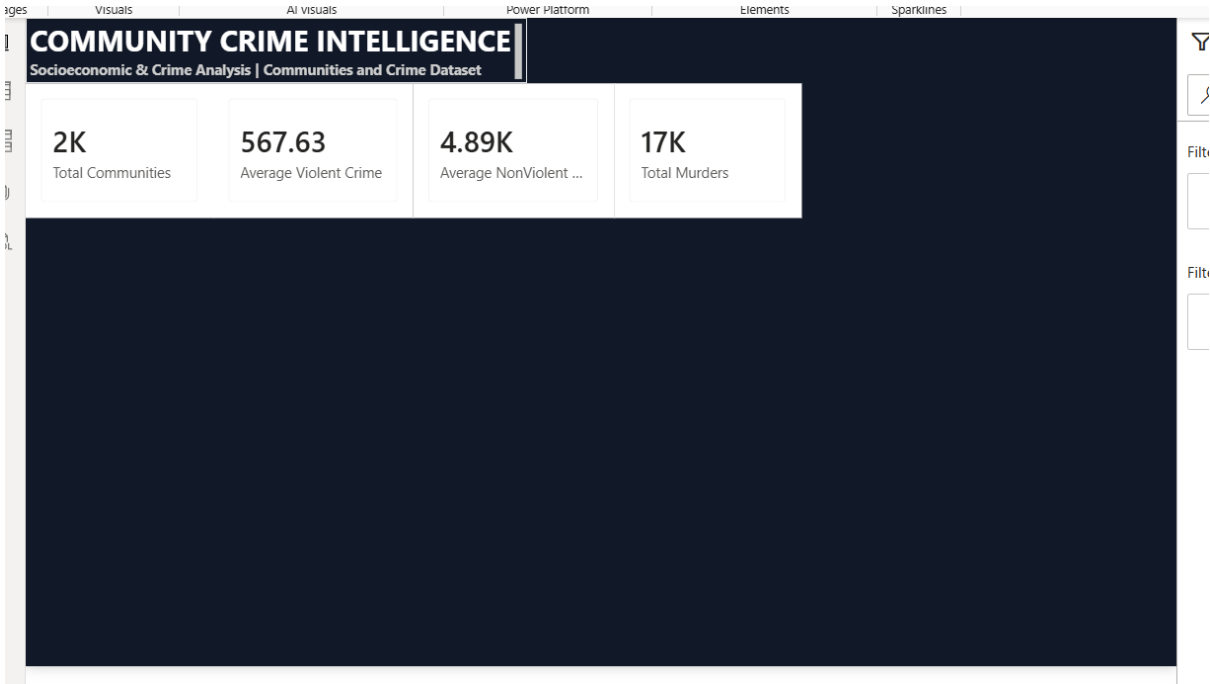
#### Total Communities



## Average Violent Crime



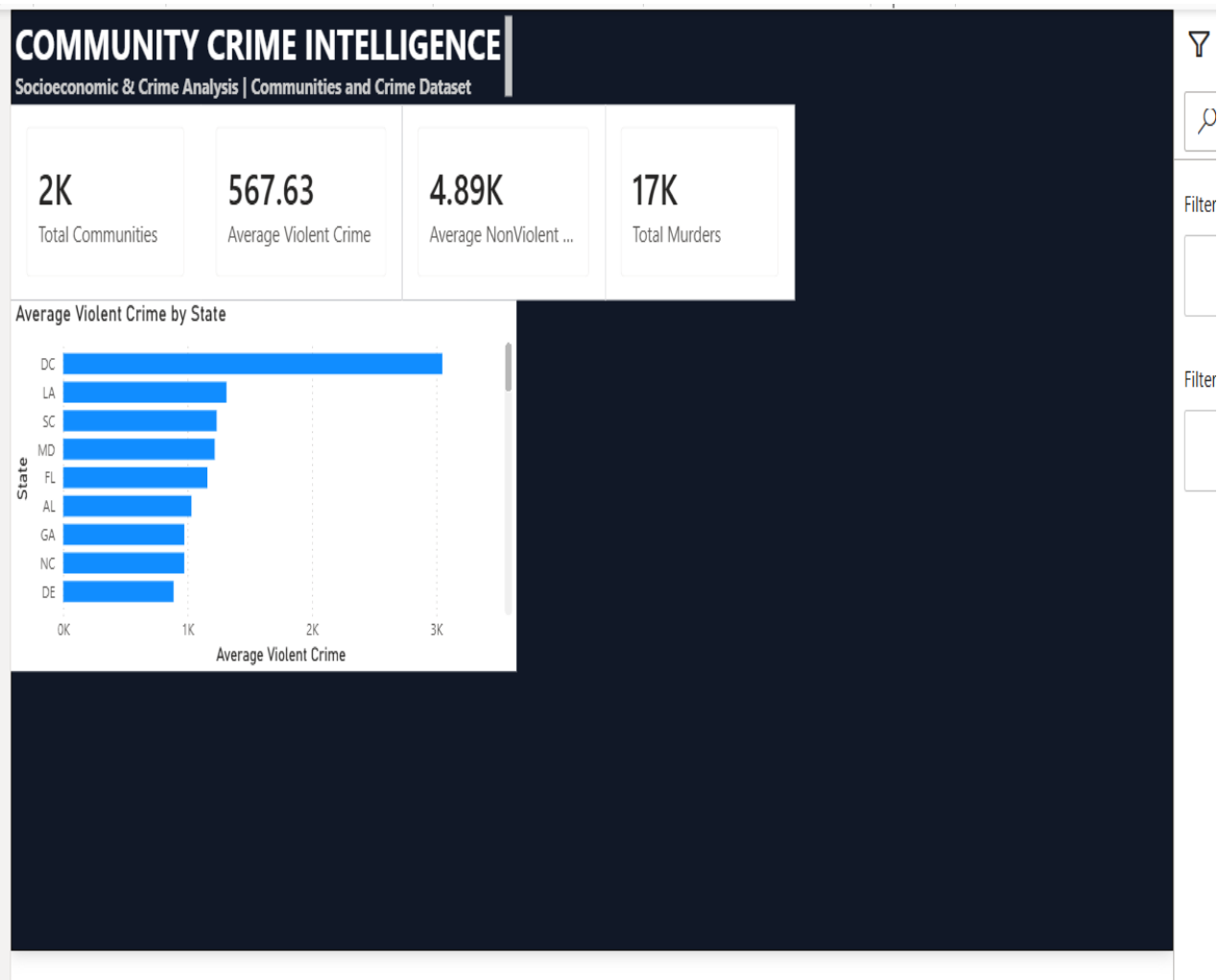
## Average Non-Violent Crime



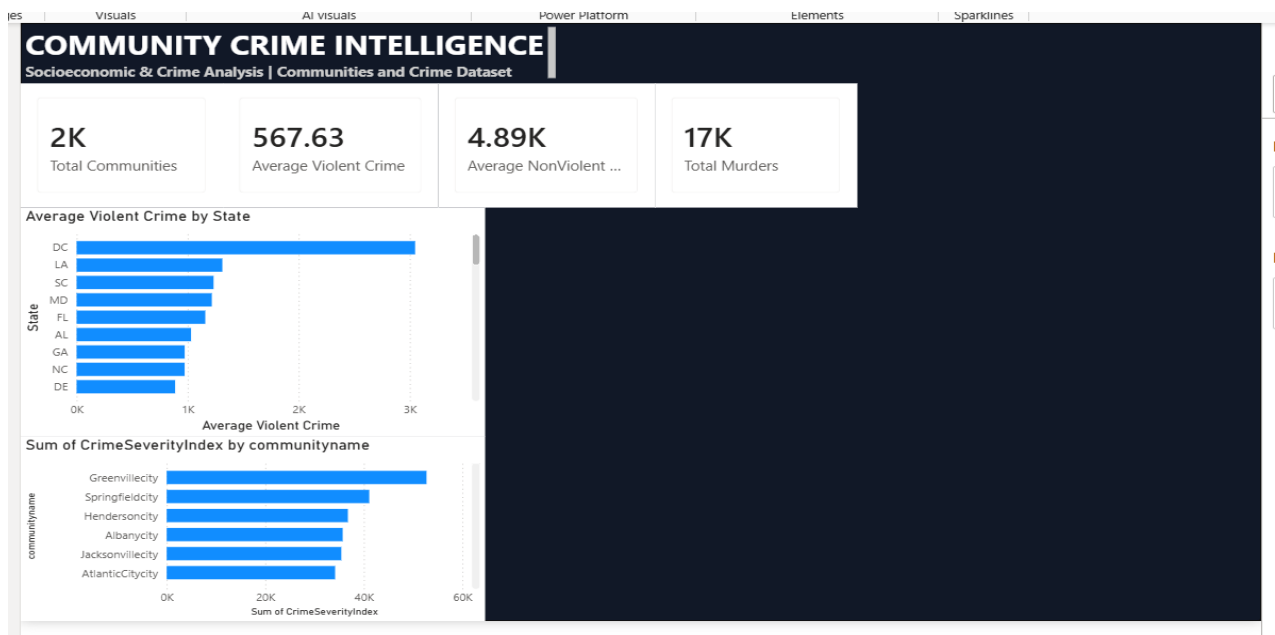
## Total Murders



## Crime by State

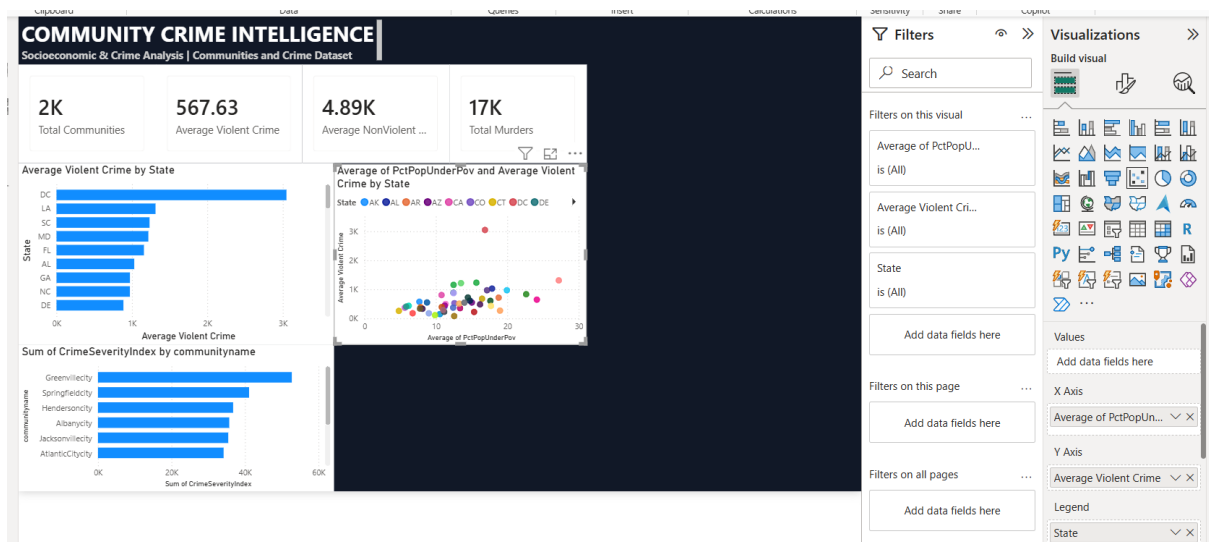


## Crime Severity by Community

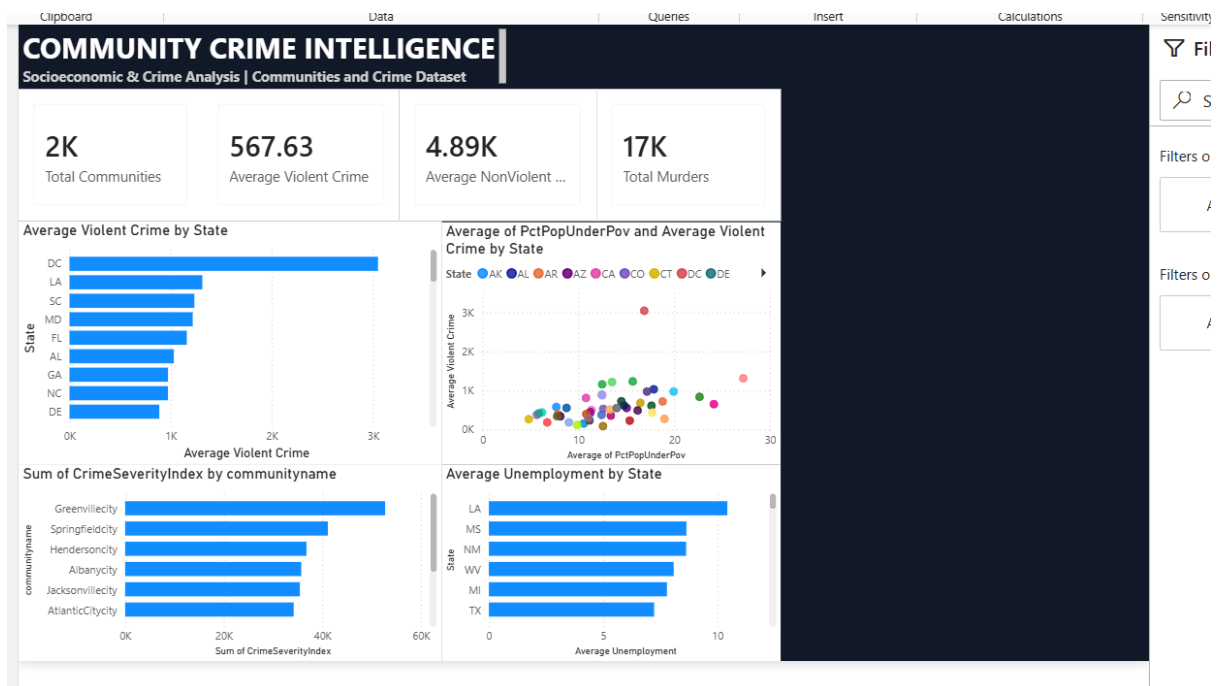




## Poverty vs Violent Crime



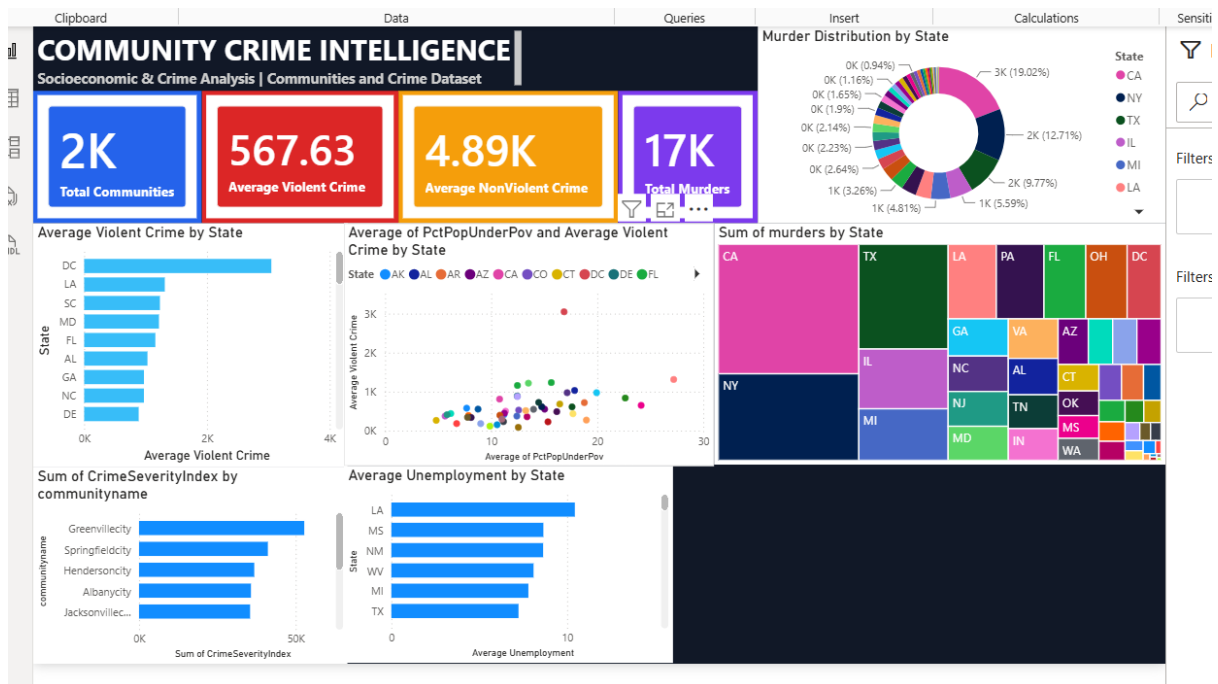
## Average Unemployment by State



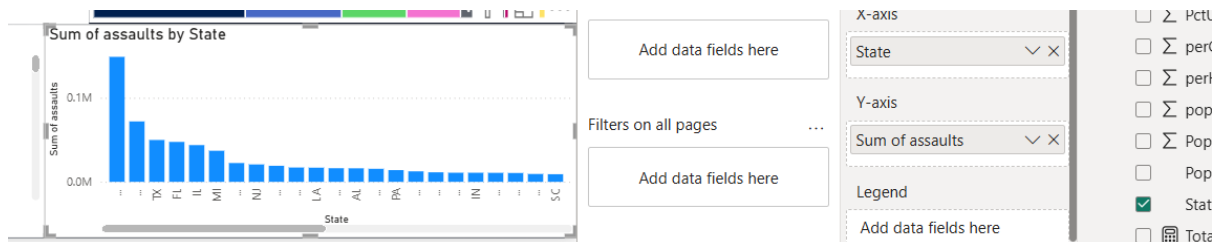
## Total Murders by State



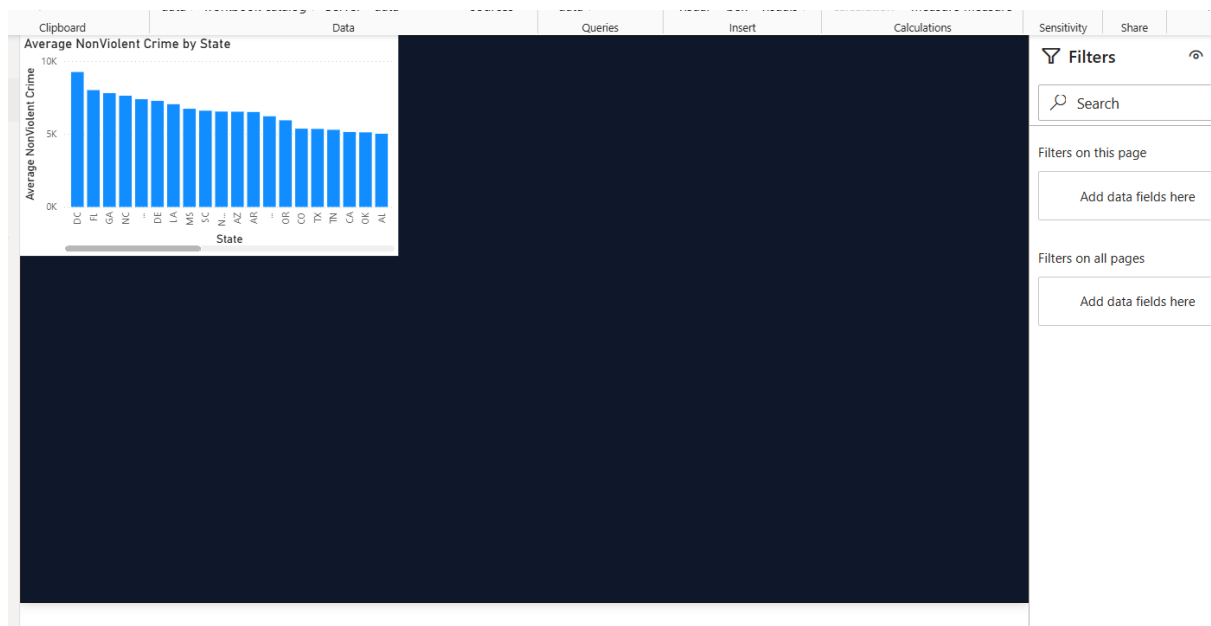
## Crime by Type



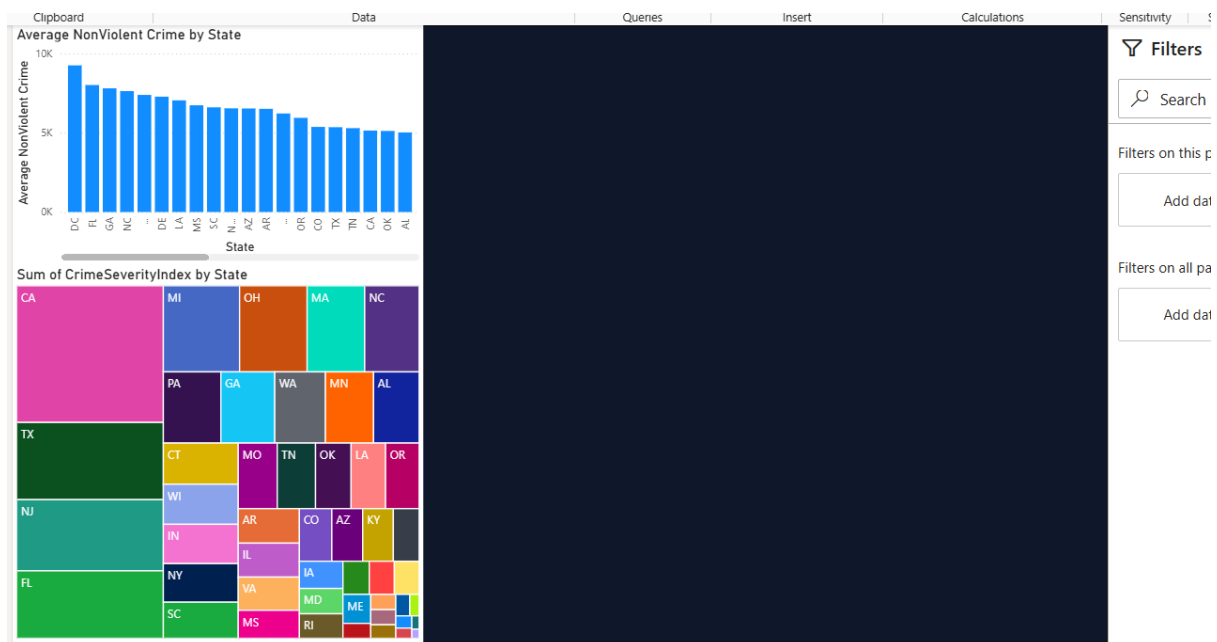
## Total Assaults by State



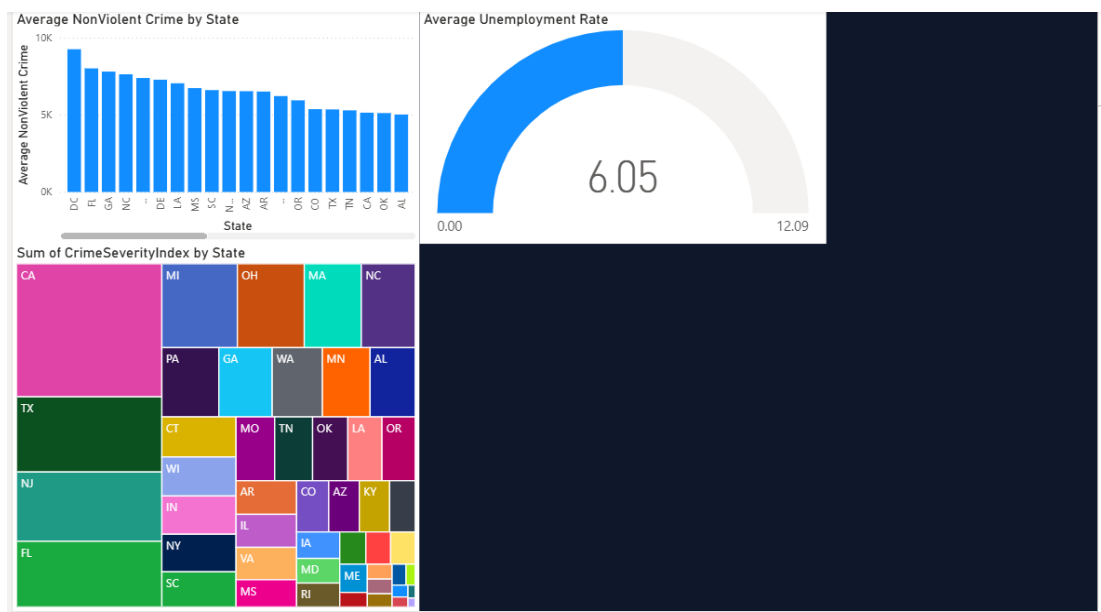
## Average NonViolent Crime by State



## Crime Severity by State



## Average Unemployment



## 9. Insights & Conclusions

The analysis of the Communities and Crime dataset identified differences in crime levels across states and communities. The dashboard helped analyze the relationship between crime and socioeconomic factors such as poverty and unemployment.

- Descriptive: Crime levels, murders, assaults, poverty, and unemployment were summarized using Power BI visuals.
- Diagnostic: Differences in crime levels were compared with socioeconomic factors to understand possible relationships.
- Predictive: High-risk communities can potentially be identified using crime severity, poverty, unemployment, and population factors.
- Prescriptive: Crime-prevention resources and support can be prioritized for communities showing higher crime severity and socioeconomic challenges.

## 10. Conclusion

The integration of **Excel and Power BI** provided an effective end-to-end approach for cleaning, modelling, analysing, and visualizing the Communities and Crime dataset. The final dashboard presents key crime and socioeconomic indicators clearly and supports **data-driven decision-making and better identification of high-risk communities**.